

In multicellular organisms, stem cells are undifferentiated cells that maintain the ability to divide indefinitely, often in an asymmetric division. One daughter cell maintains the stem cell line (self-renewal) and the other daughter cell is more differentiated and performs specialized roles or functions. These more differentiated daughter cells can also serve as progenitor cells that divide to form even further differentiated cells. During the progression from stem cell to more differentiated cells, complex changes in gene expression and metabolic profiles must occur. Endogenous microRNA-mediated RNA interference can post-transcriptionally regulate the mRNA and protein expression levels of multiple genes, making the examination of microRNAs particularly important in the exploration of stem cell differentiation.

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Which of the following conclusions is best supported by the information given in the passage?

- ☐ A. Differentiated cells contain less total mRNA content than stem cells.

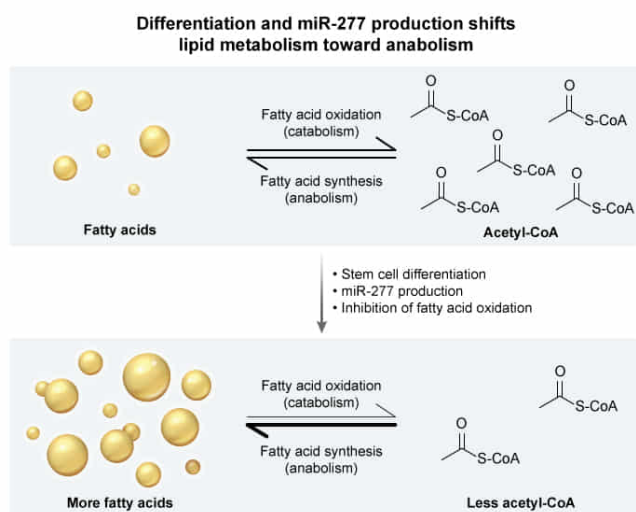
[21%]

- ✔ ☒ B. Division and differentiation of enteroblasts to enterocytes involves a shift from catabolic to anabolic lipid metabolism. [30%]
- ☐ C. Stem cells preferentially utilize glycolysis over fatty acid oxidation for energy production. [19%]
- ☐ D. As cells differentiate, miRNA-277 inhibition of some target genes is opposed by increased histone acetylation of other target genes. [28%]

Correct

30% answered correctly

Explanation:



Metabolism is broadly categorized into two branches: **anabolism and catabolism**. Anabolic reactions build up larger macromolecules from smaller precursors (eg, the synthesis of fatty acids from acetyl-CoA); these reactions typically consume energy. In contrast, catabolic reactions break down larger macromolecules into smaller molecules. Catabolic reactions generally release and capture energy—typically in the form of ATP—although energy capture is rarely 100% efficient.

Anabolic and catabolic pathways must be carefully balanced according to the energetic and material needs of the cell. When anabolic and catabolic lipid metabolism occur at the same rate simultaneously, fatty acids are built up only to be immediately broken down, leading to wasted energy with no material gain. This is called a **futile cycle**.

Therefore, active or **proliferating cells** usually favor one process over the other. The passage states that proliferating stem cells increase miRNA-277

production, leading to a decrease in fatty acid oxidation and therefore to a decrease in fatty acid catabolism. Because the fatty acids are *not* being broken down to release energy, this indicates that a **shift toward anabolic lipid metabolism** has occurred and fatty acids are most likely needed for material purposes like membrane biosynthesis.

(Choice A) Although miR-277 targets for degradation the mRNA that encodes fatty acid oxidation enzymes, many other mRNAs and many other pathways exist in the cell. The mRNA levels for genes involved in other pathways may increase in response to proliferation and differentiation; therefore, no conclusion can be drawn about total mRNA levels.

(Choice C) The passage states that miR-277 increases and fatty acid oxidation decreases for *differentiated* cells. Therefore, in *undifferentiated stem cells*, fatty acid oxidation is *not* inhibited and no conclusions can be drawn regarding the favorability of glycolysis versus fatty acid oxidation in stem cells.

(Choice D) The passage states that the inhibition of fatty acid oxidation also leads to a *decrease* in acetyl-CoA, which is needed for histone acetylation. Therefore, *increased* acetylation is unlikely.

Educational objective:

Anabolic and catabolic metabolism are balanced according to the energetic and material needs of a cell. A decrease in catabolism indicates that the metabolic equilibrium has shifted toward anabolism, and vice versa.

Time spent:0 sec

Subject:
Biochemistry

QID:403670

System:
Metabolic Reactions

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Based on the passage, miR-277 most likely also targets:

- ☐ A. pyruvate dehydrogenase. [15%]
- ☐ B. acetyl-CoA carboxylase. [27%]

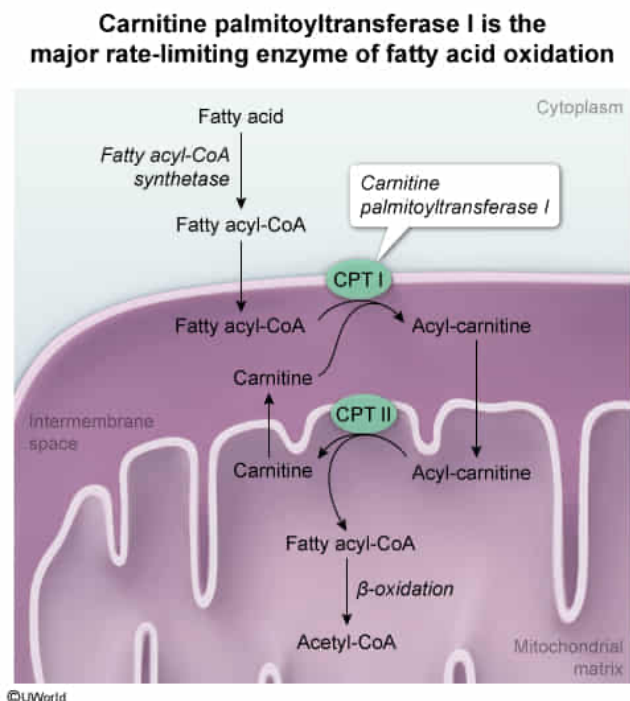
✔ ☒ C. carnitine palmitoyltransferase I. [48%]

☐ D. β -ketoacyl-ACP synthase. [7%]

Correct

49% answered correctly

Explanation:



Fatty acid oxidation is the process by which fatty acids enter the mitochondria and are broken down to release energy. **β -Oxidation** is the repetitive, four-step component of this pathway that breaks down fatty acyl-CoA molecules two carbons at a time forming acetyl-CoA molecules, which subsequently enter the **citric acid cycle**. In eukaryotes, β -oxidation occurs in the **mitochondrial matrix**; however, the **cytosolic** activation of fatty acids with coenzyme A and the transport of acyl-CoA to the matrix are important components of the fatty acid oxidation pathway as well.

Before being further catabolized, cytosolic fatty acids released from phospholipids or triglycerides are first activated by the enzyme fatty acyl-CoA synthetase, which links the acyl group onto coenzyme A to form **acyl-CoA**. Cytosolic acyl-CoA is then transported into the matrix by the **carnitine shuttle**. **Carnitine palmitoyltransferase I** (CPTI, also called carnitine acyltransferase I or CATI) transports acyl-CoA across the outer mitochondrial membrane and exchanges coenzyme A with a carnitine. A translocase antiports acylcarnitine across the inner membrane **into the matrix**, where carnitine palmitoyltransferase II replaces carnitine with coenzyme A. Free carnitine re-

enters the intermembrane space by the same translocase antiporter, continuing the shuttle cycle.

The passage states that miR-277 targets fatty acid oxidation enzymes, including acyl-CoA dehydrogenase and acetyl-CoA acyltransferase. CPTI, the enzyme that brings the fatty acid across the outer mitochondrial membrane, is considered the **rate-limiting enzyme** of fatty acid oxidation. Therefore, miR-277, which targets mRNA of enzymes of fatty acid oxidation, likely **targets CPTI** as well.

(Choice A) Pyruvate dehydrogenase catalyzes the catabolism of pyruvate to acetyl-CoA. It is extremely important in glucose oxidation but is not directly involved in fatty acid oxidation.

(Choices B and D) Acetyl-CoA carboxylase and β -ketoacyl-ACP synthase are enzymes in the fatty acid **synthesis** pathway, not the *oxidation* pathway. Furthermore, targeting these enzymes will likely lead to an *increase* in fatty acid oxidation, which is the opposite effect of miR-277 based on the passage.

Educational objective:

Carnitine palmitoyltransferase I (CPTI) is the rate-limiting enzyme of fatty acid oxidation. A microRNA that targets and inhibits fatty acid oxidation enzymes would likely target CPTI as well.

Time spent:0 sec

Subject:
Biochemistry

QID:403671

System:
Metabolic Reactions

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Administration of which hormone is most likely to replicate the effects of miR-277 on fatty acid oxidation in mammalian cells?

- ☐ A. Cortisol [8%]

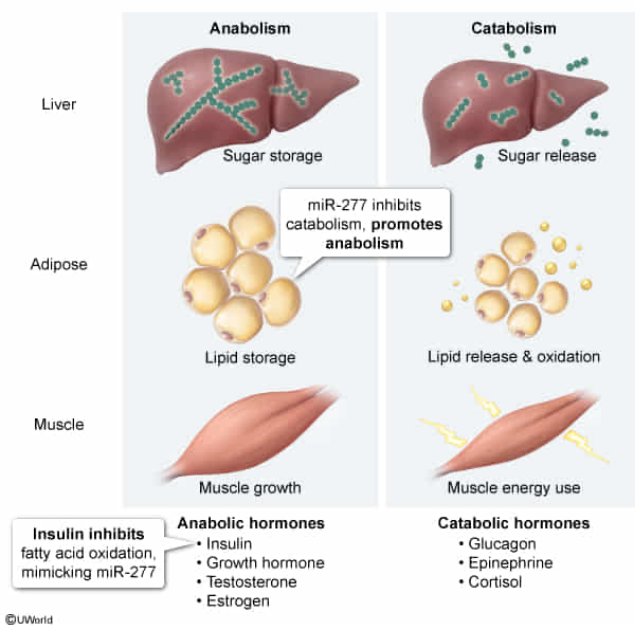
- ☐ B. Epinephrine [5%]
- ☐ C. Glucagon [22%]
- ✔ ☒ D. Insulin [63%]

Correct

62% answered correctly

Explanation:

Insulin is an anabolic hormone that inhibits fatty acid oxidation



Hormones are molecules that interact with cells and tissues within an organism to signal the activation or inhibition of various biochemical pathways.

Hormones can have various effects on different pathways within a target cell, and a single hormone may also cause different effects on the same pathway for cells in two different tissues. Nevertheless, many hormones in mammals can be broadly classified as either **anabolic** or **catabolic** based on their overall systemic effect on the metabolism of most cells.

Catabolic hormones are those that generally **stimulate** catabolic processes such as **fatty acid oxidation** in most cells they act upon.

- Glucagon is released by the α cells of the pancreas in response to **low blood glucose levels**. Glucagon stimulates fatty acid oxidation to provide energy during periods of food scarcity (**Choice C**).
- Epinephrine, also known as adrenaline, is released by the adrenal glands as part of the "fight-or-flight" response. Epinephrine stimulates fatty acid oxidation to produce energy in response to an anticipated energy need

(Choice B).

- Cortisol is a stress hormone also produced in the adrenal gland. Similar to other catabolic hormones, acute increases in cortisol increase gluconeogenesis and fatty acid oxidation in anticipation of energy needs **(Choice A)**, although chronic gluconeogenesis may promote lipogenesis by stimulating insulin production.

Anabolic hormones are those that generally stimulate anabolic processes and **inhibit catabolic processes** such as **fatty acid oxidation** in most cells they

act upon. **Insulin** is one of the most important anabolic hormones and is released by the β cells of the pancreas in response to **high blood glucose levels**. High blood glucose levels generally indicate plentiful food availability; therefore, insulin **inhibits fatty acid oxidation** and stimulates fatty acid synthesis to store metabolic fuel during periods of food abundance.

Because **miR-277** acts to **inhibit fatty acid oxidation**, a hormone that also inhibits fatty acid oxidation is needed to replicate miR-277 activity. Of the options given, **insulin** is most likely to perform this function.

Educational objective:

Inhibition of fatty acid oxidation is indicative of an anabolic metabolic state. Insulin is an anabolic hormone that inhibits fatty acid oxidation.

Time spent:0 sec

Subject:

Biochemistry

QID:403672

System:

Metabolic Reactions

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β -oxidation of saturated fatty acids involves all the following molecules EXCEPT:

- ☐ A. FADH_2 . [17%]
- ☒ B. NADPH. [53%]

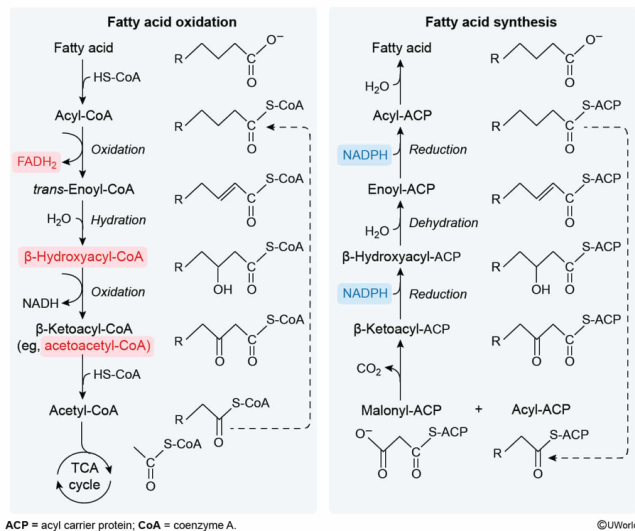
- ☐ C. acetoacetyl-CoA. [14%]
- ☐ D. β -hydroxyacyl-CoA. [14%]

Correct

53% answered correctly

Explanation:

NADPH is used in fatty acid synthesis, not oxidation



β -Oxidation is the repetitive four-step pathway that breaks down fatty acyl-CoA molecules two carbons at a time to form acetyl-CoA molecules, which can subsequently enter the **citric acid cycle**. The oxidation of a saturated acyl-CoA proceeds according to the following steps:

1. **Oxidation** of the C2–C3 bond of acyl-CoA by the enzyme **acyl-CoA dehydrogenase**, forming the α,β -unsaturated *trans*-enoyl-CoA. FAD is used as an **oxidizing agent** and is reduced to produce **FADH_2 (Choice A)**.
2. **Hydration** across the double bond by the enzyme **enoyl-CoA hydratase**. The hydration reaction stereospecifically adds the hydroxyl group to C3, the β -carbon, forming **β -hydroxyacyl-CoA (Choice D)**.
3. **Oxidation** of the hydroxyl group by the enzyme **β -hydroxyacyl-CoA dehydrogenase**. NAD^+ is used as the oxidizing agent, producing $\text{NADH} + \text{H}^+$ and a β -ketoacyl-CoA. The four-carbon β -ketoacyl-CoA formed during the final round of β -oxidation is commonly called **acetoacetyl-CoA (Choice C)**, which serves as a precursor molecule in **ketone body synthesis** and **cholesterol synthesis**.
4. **Cleavage** of the α – β bond by free coenzyme A, catalyzed by the enzyme

acetyl-CoA acyltransferase. C1 and the α -carbon leave as **acetyl-CoA**, and the oxidized β -carbon is now C1 of the **shortened acyl-CoA chain**. This reaction is reversible and can also produce acetoacetyl-CoA from two acetyl-CoA molecules during ketone body or cholesterol synthesis.

The question asks for the molecule that is *not* involved in the β -oxidation of a saturated fatty acid. **NADPH** is produced during the **pentose phosphate pathway** and is used during **fatty acid biosynthesis** but is **not involved in β -oxidation** of saturated fatty acids.

Educational objective:

β -Oxidation is a four-step repetitive pathway that breaks down acyl-CoA into acetyl-CoA and an acyl-CoA chain shortened by two carbons. β -Oxidation of saturated fatty acids produces both NADH and FADH_2 but does not include NADPH.

Time spent:0 sec

Subject:
Biochemistry

QID:403673

System:
Metabolic Reactions

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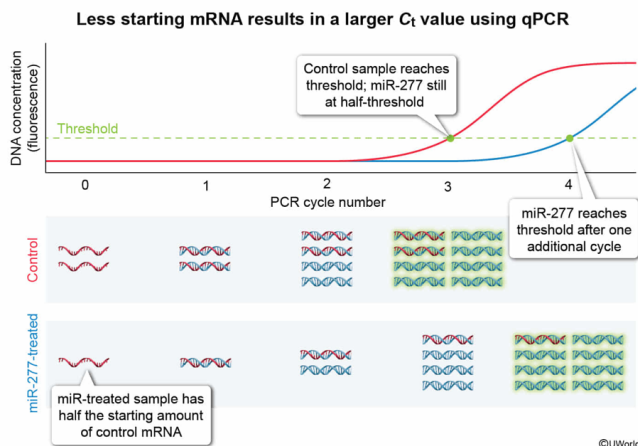
If miR-277 reduces acyl-CoA dehydrogenase mRNA levels to 50% of control levels, which of the following results is most consistent with the expected qPCR result? (Note: C_t is the threshold cycle, the number of PCR cycles required to produce a prespecified amount of cDNA.)

- ☐ A. miR-277 results in a C_t value that is half the number of cycles needed in control experiments. [28%]
- ✓ ☒ B. miR-277 results in a C_t value that is one cycle more than the number of cycles needed in control experiments. [40%]
- ☐ C. miR-277 results in a C_t value that is two cycles more than the number of cycles needed in control experiments. [27%]
- ☐ D. miR-277 causes C_t to decrease by an unpredictable amount. [3%]

Correct

39% answered correctly

Explanation:



Quantitative polymerase chain reaction (qPCR) is a technique that can be used to **quantify relative amounts of mRNA** from a sample. Isolated mRNA is first reverse-transcribed to form complementary **cDNA**, and the cDNA is then amplified by **PCR** to produce a high copy number of double-stranded amplified DNA (amplicons). Typically, a fluorescent reporter is used to monitor double-stranded amplicon production, and the number of PCR cycles required to reach a specified threshold fluorescence value is recorded. This cycle number, or C_t value, can then be related to the starting amount of RNA.

Each PCR cycle amplifies the double-stranded DNA by taking advantage of heat-stable enzymes that perform **semiconservative replication** of DNA. From a given DNA duplex, the two strands of DNA separate and each strand serves as the template to form a new duplex. In this way, each PCR cycle **doubles** the amount of DNA in a sample.

The question states that miR-277 causes acyl-CoA dehydrogenase mRNA levels

to drop to **50%** of control levels. Once enough PCR cycles have passed for the control sample to reach the threshold fluorescence, the miR-277 levels will still be at **50%** of the threshold. **One more cycle** is required to double the miR-277–treated sample's DNA levels to reach 100% of the threshold.

(Choices A and D) A decrease in C_t value indicates that *fewer* PCR cycles were needed to reach a threshold amount of dsDNA amplicons. This implies that *more* starting mRNA must have been present in the treated sample; however, miR-277 is stated to *reduce* the amount of mRNA for its targets.

(Choice C) An increase of the C_t value by 2 indicates that miR-277–treated samples needed to increase by *four times* (ie, 2×2) to match the levels of the control sample. This indicates a *four-fold* decrease in initial mRNA levels, or 25% of control levels, rather than the 50% stated in the question.

Educational objective:

Quantitative PCR is an experimental technique that measures relative mRNA levels of samples by reporting the number of cycles needed to reach a threshold amount of amplified DNA. Each 50% decrease in starting amounts of RNA requires one extra PCR cycle to reach that threshold value.

Time spent:0 sec

Subject:
Biochemistry

QID:403674

System:
Metabolic Reactions

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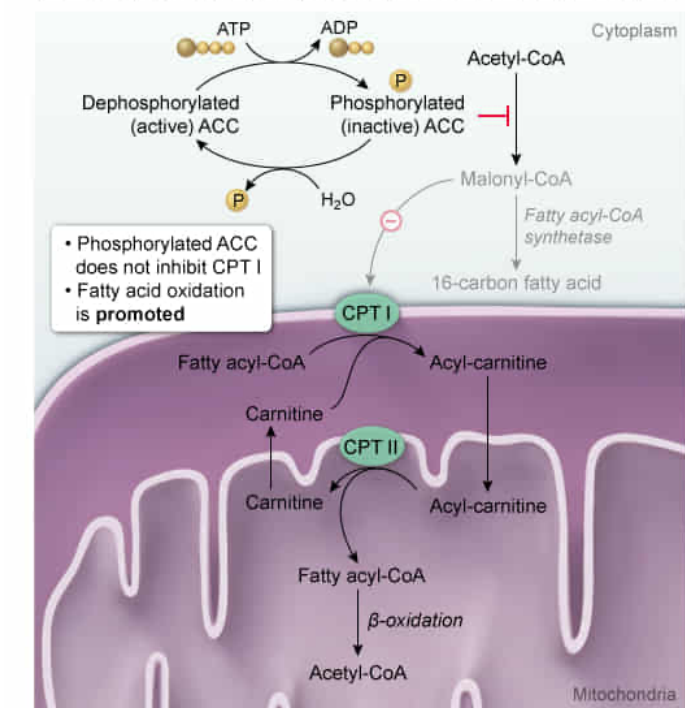
Acetyl-CoA carboxylase (ACC) is an enzyme that converts acetyl-CoA to malonyl-CoA. Phosphorylation of ACC inactivates the enzyme. Based on this, how would the effect of phosphorylating ACC most likely compare to the effects of miR-277?

- ✔ ☒ A. The ACC mutation opposes the effect of miR-277 on fatty acid oxidation. [36%]
- ☐ B. The ACC mutation has a similar effect as miR-277 on fatty acid oxidation. [38%]
- ☐ C. Fatty acid oxidation is unaffected, but the ACC mutation will deplete citric acid cycle substrates similar to miR-277. [15%]
- ☐ D. The ACC mutation produces fatty acid oxidation substrates that will accumulate if co-expressed with miR-277. [9%]

Correct
37% answered correctly

Explanation:

Acetyl-CoA carboxylase (ACC) phosphorylation promotes fatty acid oxidation, opposing miR-mediated inhibition



The regulation of a cell's metabolism can be accomplished through various mechanisms. The passage states that **miR-277 inhibits** fatty acid oxidation by targeting the mRNA of fatty acid oxidation genes, preventing new fatty acid oxidation enzymes from being translated. However, enzymes of lipid metabolism that have already been translated can be regulated by **allosteric** or **post-translational** mechanisms as well.

Acetyl-CoA carboxylase (ACC) is directly involved in **fatty acid synthesis** and carboxylates acetyl-CoA to form **malonyl-CoA**. Malonyl-CoA not only serves as a substrate for fatty acid synthase but also serves as an **allosteric regulator** of fatty acid oxidation by inhibiting **carnitine palmitoyltransferase I (CPTI)**, the rate-limiting enzyme of fatty acid oxidation. Therefore, activation of ACC inhibits fatty acid oxidation whereas **inactivation of ACC promotes fatty acid oxidation** by relieving that inhibition.

The question states that acetyl-CoA carboxylase (ACC) **phosphorylation inactivates the enzyme**. Phosphorylation of ACC can occur through the enzymes **protein kinase A** (PKA, activated through catabolic hormones like glucagon or epinephrine) or **AMP-activated protein kinase** (AMPK, activated by high AMP/low ATP levels). In both cases, these kinases signal a need for lipid catabolism to produce more ATP. Therefore, **phosphorylation of ACC promotes fatty acid oxidation**, which **opposes miR-277-mediated inhibition** of fatty acid oxidation.

(Choice B) Phosphorylation of ACC will *not* have a similar effect as miR-277 because phosphorylated ACC promotes fatty acid oxidation whereas miR-277 inhibits fatty acid oxidation.

(Choices C and D) Although ACC is not directly involved as an enzyme in the fatty acid oxidation pathway, its product (malonyl-CoA) acts as an allosteric inhibitor of fatty acid oxidation. Furthermore, phosphorylation of ACC will likely *decrease* fatty acid levels by inducing fatty acid oxidation. This will *increase* the levels of acetyl-CoA, a citric acid cycle substrate.

Educational objective:

Acetyl-CoA carboxylase (ACC) is an enzyme of fatty acid synthesis but also regulates fatty acid oxidation through its product, malonyl-CoA.

Dephosphorylated ACC stimulates synthesis and inhibits oxidation whereas *phosphorylated ACC inhibits synthesis and promotes oxidation*.

Time spent:0 sec

Subject:
Biochemistry

QID:403675

System:
Metabolic Reactions